UNIVERSIDAD NACIONAL MAYOR DE SAN MARCOS



Doctoral Program in Deep Tech, Artificial Intelligence & Emerging Technologies

Research Methods and Scientific Integrity in AI and Advanced Technologies

DELIVERABLE No. 7: Model Card & Datasheet

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This submission is **Deliverable 7**. It contains the technical documentation for the classification model and for the dataset used in my experiment. The scope, structure, and content of the documents are explained in the `README.md` file of the repository.

Aegis Content-Safety Input Guardrail — Model Card & Datasheet

This repository contains an end-to-end machine learning exercise. The goal is to classify texts as *safe* or *unsafe* using a lightweight TF-IDF + logistic regression classifier, equivalent to the "fast first layer" of a two-layer input guardrail (Moraes et al., 2026). The model is trained and evaluated on the *Aegis AI Content Safety Dataset 1.0* (NVIDIA; Ghosh et al., 2024). It is an educational documentation exercise: it is not a production safety filter and not a jailbreak detector.

The repository includes the *notebooks* of the experiment (v2, in Spanish and in English) and the two documentation files required by the course, also in both languages:

Document	Reference standard	Sections	File
Model Card	Mitchell et al. (2019)	9	`model_card_V2_ESP.md` `model_card_V2_ENG.md`
Datasheet for Datasets	Gebru et al. (2021)	7	`datasheet_V2_ESP.md` `datasheet_V2_ENG.md`

Repository contents

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├── README.md
├── model_card_V2_ESP.md
├── model_card_V2_ENG.md
├── datasheet_V2_ESP.md
├── datasheet_V2_ENG.md
├── Ejercicios_4_5_Guardrail_Aegis_ESP_v2.ipynb
└── Ejercicios_4_5_Guardrail_Aegis_ENG_v2.ipynb
```

File descriptions

- `README.md`: gives an overview of the project, its goals, its main results, and how the repository is organized.
- `model_card_V2_ESP.md / model_card_V2_ENG.md`: explain what the model is meant for, how it was evaluated, its limitations, and other important details, following the Model Card framework by Mitchell et al. (2019).
- `datasheet_V2_ESP.md / datasheet_V2_ENG.md`: describe what is in the Aegis dataset, how and why it was collected, how it should be used, and its limitations, following the framework by Gebru et al. (2021).
- `Ejercicios_4_5_Guardrail_Aegis_ESP_v2.ipynb / Ejercicios_4_5_Guardrail_Aegis_ENG_v2.ipynb`: the *notebook* I used to run the experiment (5 seeds) and generate both documents, in Spanish and in English respectively.

Key results and limits

Results (5 seeds, official *test* split of 1,199 texts): macro-F1 0.698 ± 0.008; F1 of the *unsafe* class 0.849 ± 0.002; AUPRC 0.912 ± 0.002; per-class recall: *unsafe* 0.962 and *safe* 0.406 (majority-class accuracy for reference: 0.661).

Limits: the evaluation is in-distribution only (English text, red-teaming data), so it is not directly comparable with the *zero-shot* guardrails of the big-qa-architecture repository (e.g., F1 *unsafe* of 0.6287 for the BERT → Llama-Guard chain). Aegis labels harmful content, not jailbreak techniques. The Datasheet marks as

"UNKNOWN" what the sources do not document (e.g., ethics approval and annotation period) and detects a discrepancy between the repository code comment (labels_0..labels_3) and the dataset (5 annotation columns).

Purpose of the documentation

The Model Card and the Datasheet go together with the predictive model. They record what the model and the data are like, what they should be used for, how the model was evaluated, and what to be careful about in the data. Together, they help make the work more transparent, more reproducible, and more responsible.

References

Mitchell, M., et al. (2019). *Model Cards for Model Reporting*. *Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT*)*. https://doi.org/10.1145/3287560.3287596

Gebru, T., et al. (2021). *Datasheets for Datasets*. *Communications of the ACM*, 64(12), 86–92. https://doi.org/10.1145/3458723

Ghosh, S., Varshney, P., Galinkin, E., & Parisien, C. (2024). AEGIS: Online Adaptive AI Content Safety Moderation with Ensemble of LLM Experts. arXiv:2404.05993.

Moraes, Onofre, & Aguiar (2026). Safeguarding Big Data Question-Answering Systems with a Two-Layer Input Guardrails. ICEIS 2026. https://doi.org/10.5220/0014692300004018